

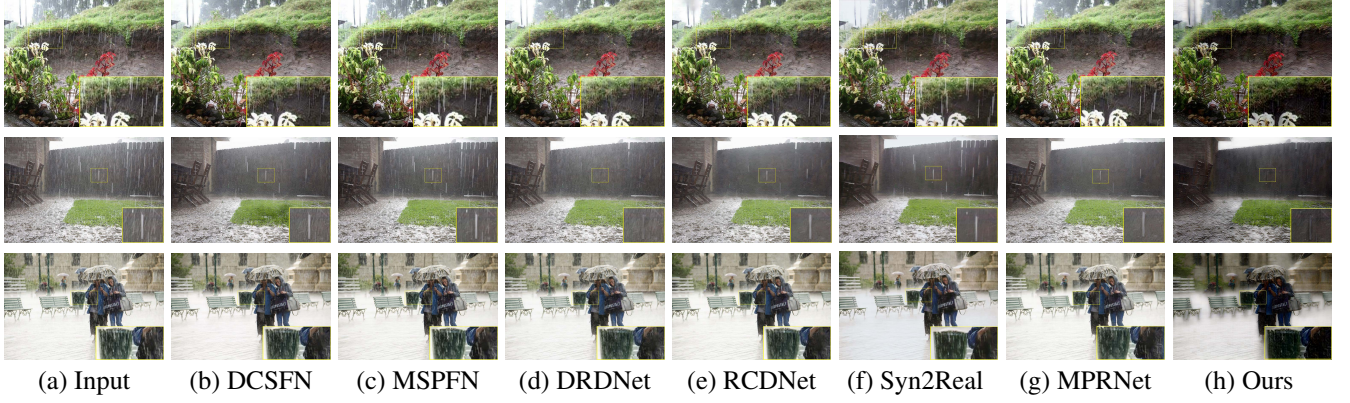


Figure 10: Comparisons with state-of-the-art methods on real-world images. Our proposed online-update learning approach is able to deal with various rainy conditions to better remove rain streaks and even haze and recover clearer images.



Figure 11: Ablation study of the proposed online-update learning approach. (a) Input. (b) Only trained on synthetic data. (c) Directly fine-tuned on real-world images. (d) Online-update learning on real-world images.



Figure 12: Applying the proposed online-update learning approach on DCSFN (?). (a) Input. (b) Only trained on synthetic data. (c) Directly fine-tuned on real-world images. (d) Online-update learning on real-world images.

truth (Fig. 11(c)), our proposed online-update learning manner (Fig. 11(d)) is able to further improve the deraining performance on real-world images, demonstrating its effectiveness.

Generality of the Online-update Learning Approach

To further demonstrate the effectiveness of the online-update learning approach for real-world image deraining, we apply it on a state-of-the-art method, DCSFN (?). Similar to Fig. 11, Fig. 12 presents the results on a real-world rainy image by different ways of learning. It can be seen that the proposed online-update learning approach can be successfully applied to DCSFN to significantly improve its deraining performance on real-world images. Since the online-update learning approach is generic, we believe it be applied to many other existing methods to improve their performance in real-world image deraining.

Conclusion

In this paper, we have proposed a high-order collaborative network with multi-scale compact constraints to control the learning process in an external and internal manner for image deraining. We have further developed a bidirectional scale-content similarity mining module to learn useful features at different scales in a down-to-up and up-to-down way to facilitate rain streaks removal. Finally, to improve the deraining performance on real-world images, we have proposed an effective online-update learning approach to fine-tune the deraining model on real-world rainy images in a self-supervised manner. Extensive experiments show that the proposed model outperforms state-of-the-art methods on five public synthetic datasets and one real-world dataset.

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